1 Fundemental Group

Definition 1.0.1

Let X be a topological space. Let $\omega, \tau : \mathbb{I} \longrightarrow X$ be two paths such that the terminal point $\omega(1)$ of ω is the same as the initial point $\tau(0)$ of τ. We then define $\omega * \tau : \mathbb{I} \longrightarrow X$ by the formula

$$\omega * \tau(t) = \begin{cases} \omega(2t), & 0 \le t \le 1/2, \\ \tau(2t-1), & 1/2 \le t \le 1. \end{cases} \tag{1.1}$$

Lemma 1.0.2

If $\omega_1 \simeq \omega_2$ and $\tau_1 \simeq \tau_2$ and $\omega_1 * \tau_1$ is defined then $\omega_2 * \tau_2$ is defined and we have, $\omega_1 * \tau_1 \simeq \omega_2 * \tau_2$.

Definition 1.0.3

We can use the standard parameterisation of any closed interval $[a, b]$ by the interval $[0, 1]$ and think of a map $\gamma : [a, b] \to X$ as a path in X, viz.,

$$\hat{\gamma}(t) := \gamma(((b-a)t + a)).$$

Theorem 1.0.4 ▶ Invariance under subdivision

Let $\gamma : [0, 1] \to X$ be a path, and $0 < t_1 < \cdots < t_n < 1$. Then γ is path homotopic to $\widehat{\gamma|_{[0,t_1]}} * \cdots * \widehat{\gamma|_{[t_n,1]}}$.

Definition 1.0.5 ▶ The Fundamental Group

Let X be any topological space and $x_0 \in X$. By the above two lemmas it follows that the set $\pi_1(X, x_0)$ consisting of all path-homotopy classes of loops in X based at x_0 forms a group under the path composition. We call this group ***the fundamental group*** of X at x_0.

Theorem 1.0.6

Let τ be a path in a space X with a, b as its initial and terminal points, respectively. Then the assignment $[\omega] \mapsto [\tau^{-1} * \omega * \tau]$ defines an isomorphism $h_{[\tau]} : \pi_1(X, a) \to \pi_1(X, b)$.

Definition 1.0.7

Given a map $f : (X, a) \longrightarrow (Y, b)$, we have a homomorphism $f_\# : \pi_1(X, a) \longrightarrow \pi_1(Y, b)$ defined by

$$f_\sharp[\omega] = [f \circ \omega]. \tag{1.2}$$

This is well defined for if $\omega \simeq \tau$ then $f \circ \omega \simeq f \circ \tau$. Since $f \circ (\omega * \tau) = (f \circ \omega) * (f \circ \tau)$, it follows that $f_\#$ is a homomorphism. Moreover, we have the following two properties which are verified directly.

1. For any space X, if $Id : X \longrightarrow X$ denotes the identity map then the induced homomorphism on the fundamental groups is the identity homomorphism, i.e., $Id_\# = Id$.
2. Given maps $f : (X, a) \longrightarrow (Y, b)$ and $g : (Y, b) \longrightarrow (Z, c)$ we have $(g \circ f)_\# = g_\# \circ f_\#$.